Fiscal Year:	FY 2023	**Task Last Updated:**	FY 09/23/2022
PI Name:	Radlinska, Aleksandra Ph.D.		
Project Title:	Microgravity Investigation of Cement Solidification (MICS)		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	University Park	**State:**	PA
Zip Code:	16802-1408	**Congressional District:**	5
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2015 NNH15ZTT002N MaterialsLab Open Science Campaigns for Experiments on the International Space Station
Start Date:	11/22/2016	**End Date:**	11/21/2022
No. of Post Docs:		**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA MSFC
Contact Monitor:	Jackson, Enrique	**Contact Phone:**	
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Flight Program:	ISS		
Flight Assignment:	ISS NOTE: End date changed to 11/21/2022 per NSSC information (Ed., 10/15/21) NOTE: End date changed to 11/21/2021 per NSSC information (Ed., 10/1/20)		
Key Personnel Changes/Previous PI:	one graduate student continues PhD studies		
COI Name (Institution):	Scheetz, Barry Ph.D. (Pennsylvania State University)		
Grant/Contract No.:	NNX17AC48G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The intent of this work is to utilize the microgravity environment aboard the International Space Station (ISS) to fully investigate the complex process of cement solidification. Microstructural development of hydrating cement occurs in stages during the hydration reaction and hardening process, which results in elaborate combinations of amorphous and crystalline phases. The morphology, volume fraction, and distribution of these phases ultimately determine the hardened cement's material properties. Minimizing gravity-driven phenomena, such as thermosolutal convective flow and sedimentation, will ensure crystal growth strictly by diffusion and a considerably different microstructure than that observed in typical laboratory conditions on Earth.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The proposed microgravity experiments will satisfy two major objectives. First, the microstructure and material properties of the benchmark samples will be evaluated; different responses to thermal and mechanical loading are expected and will be characterized in detail. Any positive attributes found in the special environment will be reported with the intent of incorporating and improving Earth-based cement and concrete processing. Given the magnitude of cement usage worldwide, even a slight change in manufacturing or usage can have huge implications on the sustainability of the cement industry.
Task Progress:	Task 1 – Analysis of the Techshot Multi-use Variable-gravity Platform (MVP) Samples The Microgravity Investigation of Cement Solidification – Multi-use Variable-gravity Platform (MICS-MVP) experiment was able to mimic the gravity levels of the Moon (0.17g), Mars (0.38g), and a third statistical data point at 7/10th of Earth's gravity. Samples were also mixed and left in the microgravity environment on the International Space Station (ISS) and a sample was mixed on Earth for a total of five gravity levels. A deeper understanding of the micromechanical properties and microstructure was developed on the ordinary portland cement with lunar regolith simulant, Johnson Space Center 1-A (JSC-1A), mortars. 1.1. Analysis of the Ordinary Portland Cement with Lunar Simulant Mortars – Nanoindentation was conducted on the samples to understand if there are any differences in the micromechanical properties of the samples as a function of gravity. The intent is to understand if any variations exist and how they would influence the macroscale strength of the sample. Results showed that there is no statistical difference in the micromechanical properties as a function of the gravity level and that other microstructure features, such as the increase in the amount of trapped air as the gravity level is decreased, would have a more drastic influence on the strength. Task 2 – Supplementary Ground Research Building off Task 1, in-situ resource utilization must be kept in mind for progressing concrete materials for lunar infrastructure. The development of a geopolymer lunar concrete material is ongoing and showing positive signs. 2.1. Development of an In-Situ Lunar Concrete – The geopolymer lunar concrete technology is promising for creating fundamental and necessary lunar infrastructure. This past year, research activities focused on curing, and exposing the material to environmental conditions expected on the lunar surface were explored. Samples were cured at reduced atmospheric pressures and tested for compressive strength. Samples were also placed within a vacuum chamber after curing to monitor the compressive strength as a function of exposure time. The MICS team conducted an experiment on the Materials International Space Station Experiment Flight Facility (MISSE-FF) this past year. Eight samples were placed in the Zenith direction (two ordinary portland cement pastes and six geopolymer lunar concrete samples) to understand the influence of the space environment on the samples.
Bibliography Type:	Description: (Last Updated: 10/02/2023)
Abstracts for Journals and Proceedings	Collins PJ, Edmunson J, Fiske M, Radlinska A. "Geopolymer lunar concrete design utilizing a response surface methodology." ASTM International Conference on Additive Manufacturing (ICAM), Orlando, FL, October 31 - November 04, 2022. Abstracts. ASTM International Conference on Additive Manufacturing (ICAM), Orlando, FL, October 31 - November 04, 2022. , Oct-2022
Articles in Peer-reviewed Journals	Collins PJ, Edmunson J, Fiske M, Radlinska A. "Materials characterization of various lunar regolith simulants for use in geopolymer lunar concrete." Adv Space Res. 2022, Jun 1;69(11):3941-51. https://doi.org/10.1016/j.asr.2022.03.012 , Jun-2022
Papers from Meeting Proceedings	Collins PJ, Edmunson J, Fiske JM, Bilén SG, Radlinska A. "Geopolymer lunar concrete under reduced-pressure curing and vacuum exposure." IAC 2022: 73rd International Astronautical Congress, Paris, France, September 18-22, 2022. Abstracts. IAC 2022: 73rd International Astronautical Congress, Paris, September 18-22, 2022. IAC Paper ID 69534. https://dl.iafastro.directory/event/IAC-2022/paper/69534/ , Sep-2022